

GIT ASSIGNMENT - 2

1. Install git.

git init

```
HP@Ashish MINGW64 ~/Downloads/Git (chintu)
$ git init
Reinitialized existing Git repository in C:/Users/HP/Downloads/Git/.git/

HP@Ashish MINGW64 ~/Downloads/Git (chintu)
$
```

2. Create a repo in github with README.md and .ignore file.

vi .gitignore

git add .

git commit -m "message"

```
HP@Ashish MINGW64 ~/Downloads/Git (master)
$ ls
README.md  devops  tehie  test1
```

```
HP@Ashish MINGW64 ~/Downloads/Git (release)
$ ls -a
./ ../ .git/ .gitignore a b c devops horizon index.html* tehie test1

HP@Ashish MINGW64 ~/Downloads/Git (release)
$ cat .gitignore
*.txt
sunith
```

3. Clone the created repo to local.

git clone git@github.com:Sunith3210/Git_Practice.git

```
HP@Ashish MINGW64 ~/Downloads (master)
$ git clone git@github.com:Sunith3210/Git_Practice.git
Cloning into 'Git_Practice'...
remote: Enumerating objects: 29, done.
remote: Counting objects: 100% (29/29), done.
remote: Compressing objects: 100% (15/15), done.
remote: Total 29 (delta 7), reused 20 (delta 4), pack-reused 0 (from 0)
Receiving objects: 100% (29/29), done.
Resolving deltas: 100% (7/7), done.
```

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4. Create two files in local repo.

created with touch command

touch a b

```
HP@Ashish MINGW64 ~/Downloads/Git (master)
$ touch a b

HP@Ashish MINGW64 ~/Downloads/Git (master)
$ ls
README.md  a  b  devops  tehie  test1

HP@Ashish MINGW64 ~/Downloads/Git (master)
$ git add a

HP@Ashish MINGW64 ~/Downloads/Git (master)
$ git commit -m "added in my local machine"
[master dfe8e79] added in my local machine
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 a

HP@Ashish MINGW64 ~/Downloads/Git (master)
$ git add b

HP@Ashish MINGW64 ~/Downloads/Git (master)
$ git commit -m "added in my local machine"
[master cf8a17a] added in my local machine
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 b
```

5. Commit two files and push to central Repository.

git add .

git commit -m "added in my local machine"

git push -u origin master

```
HP@Ashish MINGW64 ~/Downloads/Git (master)
$ git push -u origin master
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 512 bytes | 512.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/Sunith3210/Git_Practice.git
   535eaae..cf8a17a  master -> master
branch 'master' set up to track 'origin/master'.
```

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6. Create a branch in local and create a sample file and push to central.

git branch Tech

touch abc

git commit -m "added this for tech branch"

```
HP@Ashish MINGW64 ~/Downloads/Git (master)
$ git branch Tech

HP@Ashish MINGW64 ~/Downloads/Git (master)
$ git checkout Tech
Switched to branch 'Tech'

HP@Ashish MINGW64 ~/Downloads/Git (Tech)
$ touch abc

HP@Ashish MINGW64 ~/Downloads/Git (Tech)
$ git add abc

HP@Ashish MINGW64 ~/Downloads/Git (Tech)
$ git commit -m "this was added for tech branch"
[Tech d22790f] this was added for tech branch
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 abc

HP@Ashish MINGW64 ~/Downloads/Git (Tech)
$ git push -u origin Tech
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 234 bytes | 234.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'Tech' on GitHub by visiting:
remote:   https://github.com/Sunith3210/Git_Practice/pull/new/Tech
remote:
To https://github.com/Sunith3210/Git_Practice.git
 * [new branch]      Tech -> Tech
branch 'Tech' set up to track 'origin/Tech'.

HP@Ashish MINGW64 ~/Downloads/Git (Tech)
$ |
```

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7. Create a branch in github and clone that to local.

Created a branch in github and clone it to my existing repo

git fetch origin

git switch -c batch --track

```
HP@Ashish MINGW64 ~/Downloads/Git (master)
$ ls
Git_Practice/  README.md  a  b  devops  tehie  test1

HP@Ashish MINGW64 ~/Downloads/Git (master)
$ git fetch origin

HP@Ashish MINGW64 ~/Downloads/Git (master)
$ git switch -c batch --track origin/batch
Switched to a new branch 'batch'
branch 'batch' set up to track 'origin/batch'.

HP@Ashish MINGW64 ~/Downloads/Git (batch)
$ git branch --list
  Tech
* batch
  chintu
  hotfix
  master
  release
```

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8. Merge the created branch with master in git local.

```
vi file.txt
```

```
git add file.txt
```

```
git commit -m "commit message"
```

```
git merge tech
```

```
HP@Ashish MINGW64 ~/Downloads/Git (batch)
$ vi file.txt

HP@Ashish MINGW64 ~/Downloads/Git (batch)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches it

HP@Ashish MINGW64 ~/Downloads/Git (batch)
$ git commit -m "added in batch"
[batch 0803e33] added in batch
1 file changed, 1 insertion(+)
create mode 100644 file.txt

HP@Ashish MINGW64 ~/Downloads/Git (batch)
$ git merge tech
Auto-merging file.txt
CONFLICT (add/add): Merge conflict in file.txt
Automatic merge failed; fix conflicts and then commit the result.

HP@Ashish MINGW64 ~/Downloads/Git (batch|MERGING)
$ vi file.txt

HP@Ashish MINGW64 ~/Downloads/Git (batch|MERGING)
$ git add file.txt

HP@Ashish MINGW64 ~/Downloads/Git (batch|MERGING)
$ git commit -m "this was merged with tech batch file"
[batch 1d51326] this was merged with tech batch file

HP@Ashish MINGW64 ~/Downloads/Git (batch)
$ git push -u origin batch
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (9/9), 854 bytes | 854.00 KiB/s, done.
Total 9 (delta 3), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/Sunith3210/Git_Practice.git
cf8a17a..1d51326 batch -> batch
branch 'batch' set up to track 'origin/batch'.
```

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9. Merge the created branch with master in github by sending a pull request.

```
git branch devops
```

```
vi branch.txt
```

```
git add .
```

```
git commit
```

```
git push -u origin devops
```

```
HP@Ashish MINGW64 ~/Downloads/Git (devops)
$ git push -u origin devops
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (7/7), 695 bytes | 695.00 KiB/s, done.
Total 7 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), done.
remote:
remote: Create a pull request for 'devops' on GitHub by visiting:
remote:   https://github.com/Sunith3210/Git_Practice/pull/new/devops
remote:
To https://github.com/Sunith3210/Git_Practice.git
* [new branch]      devops -> devops
branch 'devops' set up to track 'origin/devops'.
```

 **devops** had recent pushes 42 seconds ago [Compare & pull request](#)

 **Your master branch isn't protected** [Dismiss](#) [Protect this branch](#)
Protect this branch from force pushing or deletion, or require status checks before merging. [View documentation.](#)

 **master**  **7** Branches  **0** Tags [Add file](#) [Code](#)

 **Sunith3210** added in my local machine cf8a17a · 3 hours ago  **5** Commits

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10. Create a file in local and send that to branch in github.

```
vi techie
```

```
git add .
```

```
git commit -m "added in devops branch"
```

```
git push -u origin devops
```

```
HP@Ashish MINGW64 ~/Downloads/Git (devops)
$ vi techi_horizon

HP@Ashish MINGW64 ~/Downloads/Git (devops)
$ git add techi_horizon
warning: in the working copy of 'techi_horizon', LF will be replaced by CRLF the next time you commit.

HP@Ashish MINGW64 ~/Downloads/Git (devops)
$ git commit -m "added for techie_horizon"
[devops e3190b2] added for techie_horizon
1 file changed, 1 insertion(+)
create mode 100644 techi_horizon

HP@Ashish MINGW64 ~/Downloads/Git (devops)
$ git push -u origin devops
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 322 bytes | 322.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Sunith3210/Git_Practice.git
137fb5f..e3190b2 devops -> devops
branch 'devops' set up to track 'origin/devops'.

HP@Ashish MINGW64 ~/Downloads/Git (devops)
$ |
```

devops

7 Branches 0 Tags

Go to file

Add file

Code

This branch is 3 commits ahead of and 3 commits behind master .

Sunith3210

added for techie_horizon

e3190b2 · 2 minutes ago

5 Commits

README.md	Create README.md	13 minutes ago
a	Create README.md	13 minutes ago
b	Create README.md	13 minutes ago
file.txt	this added in devops branch	9 minutes ago
techi_horizon	added for techie_horizon	2 minutes ago
tehie	this added in my local machine	5 hours ago
test1	Create test1	4 hours ago

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11.Clone only a branch from github to local.

`git clone -b <branch name> <paste the link>`

```
HP@Ashish MINGW64 ~
$ git clone -b devops git@github.com:Sunith3210/Git_Practice.git
Cloning into 'Git_Practice'...
remote: Enumerating objects: 51, done.
remote: Counting objects: 100% (51/51), done.
remote: Compressing objects: 100% (26/26), done.
remote: Total 51 (delta 16), reused 42 (delta 11), pack-reused 0 (from 0)
Receiving objects: 100% (51/51), 5.69 KiB | 1.42 MiB/s, done.
Resolving deltas: 100% (16/16), done.
```

```
HP@Ashish MINGW64 ~
$ cd Git_Practice/

HP@Ashish MINGW64 ~/Git_Practice (devops)
$ ls
README.md a b file.txt techi_horizon tehie test1

HP@Ashish MINGW64 ~/Git_Practice (devops)
$ git branch --list
* devops

HP@Ashish MINGW64 ~/Git_Practice (devops)
$ |
```

12.Create a file with all passwords and make that untrackable with git.

`echo "passwords.txt" > .gitignore`

`git add passsswords.txt` (then we can see it will ignore)

```
HP@Ashish MINGW64 ~/Downloads/Git (devops)
$ git push -u origin devops
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 288 bytes | 288.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Sunith3210/Git_Practice.git
   e3190b2..77148da  devops -> devops
branch 'devops' set up to track 'origin/devops'.

HP@Ashish MINGW64 ~/Downloads/Git (devops)
$ vi passwords.txt

HP@Ashish MINGW64 ~/Downloads/Git (devops)
$ git add passwords.txt
The following paths are ignored by one of your .gitignore files:
passwords.txt
hint: Use -f if you really want to add them.
hint: Turn this message off by running
hint: "git config advice.addIgnoredFile false"
```


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13. Make a commit and make that commit reset without saving changes.

```
echo "Hello Git" > test.txt
git add test.txt
git commit -m "Add test file"
git log --oneline
git reset --hard HEAD~1
git status
```

```
HP@Ashish MINGW64 ~/Downloads/Git (devops)
$ git log --oneline
1c824c2 (HEAD -> devops) My commit
77148da (origin/devops) this is my password
e3190b2 added for techie_horizon
137fb5f this added in devops branch
737a122 (master) Create README.md
d57e9f4 Create test1
cded1e1 this added in my local machine

HP@Ashish MINGW64 ~/Downloads/Git (devops)
$ git reset --hard HEAD~1
HEAD is now at 77148da this is my password

HP@Ashish MINGW64 ~/Downloads/Git (devops)
$ git status
On branch devops
Your branch is up to date with 'origin/devops'.

nothing to commit, working tree clean
```

14. Revert a committed commit to the older version.

Identify the commit to undo.

Run `git revert` using the commit id.

Resolve conflicts if prompted, then complete the revert.

Push the new revert commit

```
git revert <commit id>
```

```
[devops 5880614] Revert "added for techie_horizon"
1 file changed, 1 deletion(-)
delete mode 100644 techi_horizon

HP@Ashish MINGW64 ~/Downloads/Git (devops)
$ |
```

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15. Push a file to stash without saving the changes and work on another file.

```
git add .  
git stash  
git status  
git stash list
```

```
HP@Ashish MINGW64 ~/Downloads/Git (devops)  
$ git add abc  
  
HP@Ashish MINGW64 ~/Downloads/Git (devops)  
$ git status  
On branch devops  
Your branch is ahead of 'origin/devops' by 1 commit.  
  (use "git push" to publish your local commits)  
  
Changes to be committed:  
  (use "git restore --staged <file>..." to unstage)  
    new file:   abc  
  
HP@Ashish MINGW64 ~/Downloads/Git (devops)  
$ git stash  
Saved working directory and index state WIP on devops: 5880614 Revert "added for techie_horizon"  
  
HP@Ashish MINGW64 ~/Downloads/Git (devops)  
$ git status  
On branch devops  
Your branch is ahead of 'origin/devops' by 1 commit.  
  (use "git push" to publish your local commits)  
  
nothing to commit, working tree clean  
  
HP@Ashish MINGW64 ~/Downloads/Git (devops)  
$ git stash list  
stash@{0}: WIP on devops: 5880614 Revert "added for techie_horizon"
```

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16. Undo the stash file and start working on that again.

git stash list

git stash pop

```
HP@Ashish MINGW64 ~/Downloads/Git (devops)
$ git stash list
stash@{0}: WIP on devops: 5880614 Revert "added for techie_horizon"

HP@Ashish MINGW64 ~/Downloads/Git (devops)
$ git stash pop
On branch devops
Your branch is ahead of 'origin/devops' by 1 commit.
(use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   abc

Dropped refs/stash@{0} (508a5ae5f2c6809941b6ccd3f4f0bde7369c56f6)

HP@Ashish MINGW64 ~/Downloads/Git (devops)
$ |
```

17. Generate a ssh-keygen and configure into github.

ssh-keygen

login to github account

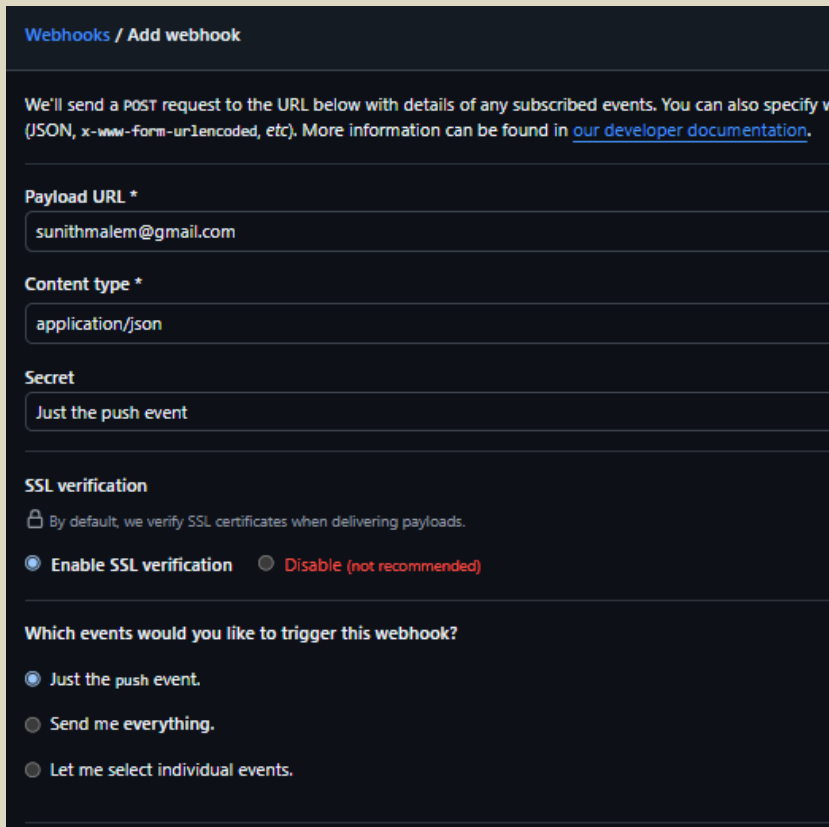
Start the SSH agent and add the private key. Copy the public key. In GitHub, open Settings → SSH and GPG keys → New SSH key and paste the public key. Test authentication. Optionally change the repository remote from HTTPS to SSH

The image shows a terminal window at the top where an SSH key is generated using the command `ssh-keygen -t rsa -b 4096 -C 'HP@Ashish' -f .ssh/id_rsa`. Below the terminal is a screenshot of the GitHub 'SSH keys' management page. The page has a dark theme and a green 'New SSH key' button in the top right. It displays a list of authentication keys. One key is shown: 'Ashish_lapi' with a SHA256 fingerprint of 'at2XsWn7MFqDCwZD7s8ABqKMRatdu6x3nigPhhn3oVQ', added on Aug 24, 2026. A 'Delete' button is visible next to the key. The page also includes a note about removing unrecognized keys and a status for the key: 'Last used within the last week — Read/write'.

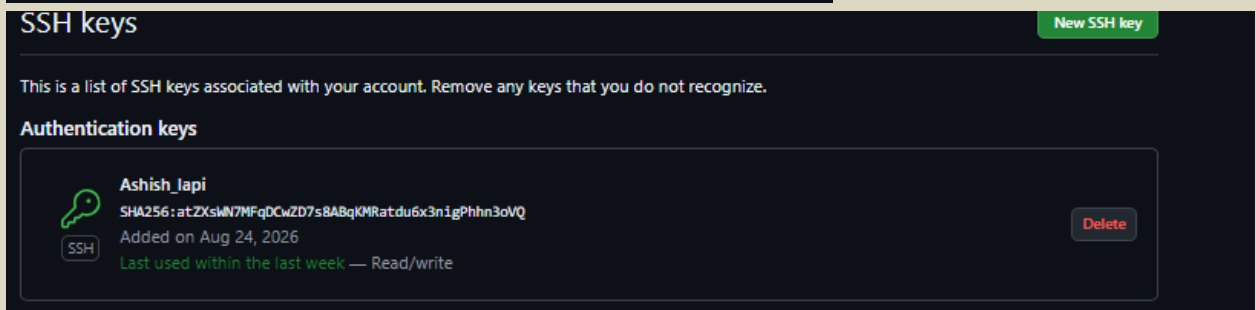
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18. Configure webhooks to github.

Create/select a GitHub repository. Go to: Repository → Settings → Webhooks → Add webhook



The screenshot shows the 'Webhooks / Add webhook' page in GitHub. It includes a description of webhooks, input fields for 'Payload URL' (suniithmalem@gmail.com), 'Content type' (application/json), and 'Secret' (Just the push event). There are radio buttons for 'SSL verification' (Enable SSL verification is selected) and 'Which events would you like to trigger this webhook?' (Just the push event is selected).



The screenshot shows the 'SSH keys' page in GitHub. It has a 'New SSH key' button in the top right. Below the heading, it says 'This is a list of SSH keys associated with your account. Remove any keys that you do not recognize.' Under the 'Authentication keys' section, there is a table with one entry: 'Ashish_lapi' with a SHA256 key, added on Aug 24, 2026, and last used within the last week. A 'Delete' button is next to the entry.

SSH key	Added on	Last used	Permissions	Action
Ashish_lapi SHA256:atZXsW7MFqDCwZD7s8ABqKMRatdu6x3nigPhhn3oVQ	Aug 24, 2026	Last used within the last week	Read/write	Delete

19. Basic understanding of .git file.

1. What is .git?

.git is a **hidden directory** created by Git inside a project when you initialize or clone a Git repository. It stores the repository's **Git metadata, configuration, commits, branches, and history**.

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20. Check all the logs of git.

`git log`

`git log --oneline`

```
HP@Ashish MINGW64 ~/Downloads/Git (batch)
$ git log
commit 1d51326d9ac55a3b79d9dfd79c6d694c4512a0e0 (HEAD -> batch, origin/batch)
Merge: 0803e33 6947389
Author: sunith <sunithmalem@gmail.com>
Date: Tue Aug 25 18:57:44 2026 +0530

    this was merged with tech batch file

commit 0803e338dfcad6e1ad9ca1c470dcd6a8cc96e221
Author: sunith <sunithmalem@gmail.com>
Date: Tue Aug 25 18:56:20 2026 +0530

    added in batch

commit 69473893b122f45671c455936401c393abd9c1e2 (tech)
Author: sunith <sunithmalem@gmail.com>
Date: Tue Aug 25 18:50:30 2026 +0530

    this is added in tech

commit d22790fb8413d89d5d59676a44e64846af4433eb (origin/Tech)
Author: sunith <sunithmalem@gmail.com>
Date: Tue Aug 25 18:15:20 2026 +0530

    this was added for tech branch

commit cf8a17a3f664658a8cf3ef3bfa2e5e1d23da8576 (origin/master, master)
Author: sunith <sunithmalem@gmail.com>
Date: Tue Aug 25 18:05:50 2026 +0530

    added in my local machine

commit dfe8e792de834d8d83a3530a805e2ed9204c1585
Author: sunith <sunithmalem@gmail.com>
Date: Tue Aug 25 18:05:39 2026 +0530

    added in my local machine

commit 535eaae4c56efc5e0f4ebc08f45086080e51703c
Author: Sunith <135949976+Sunith3210@users.noreply.github.com>
Date: Tue Aug 25 17:54:57 2026 +0530

    Create README.md
```

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21. Rename the commit message.

```
git add
```

```
git commit -m "commit message"
```

```
git commit --amend (changed)
```

```
[master 37eef67] this line is added in local machine
Date: Tue Aug 25 18:05:50 2026 +0530
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 b

HP@Ashish MINGW64 ~/Downloads/Git (master)
$ git log
commit 37eef67b8a20f8067f0fd5994fad98ad47d92b45 (HEAD -> master)
Author: sunith <sunithmalem@gmail.com>
Date: Tue Aug 25 18:05:50 2026 +0530

    this line is added in local machine

commit dfe8e792de834d8d83a3530a805e2ed9204c1585
Author: sunith <sunithmalem@gmail.com>
Date: Tue Aug 25 18:05:39 2026 +0530

    added in my local machine

commit 535eaae4c56efc5e0f4ebc08f45086080e51703c
Author: Sunith <135949976+Sunith3210@users.noreply.github.com>
Date: Tue Aug 25 17:54:57 2026 +0530

    Create README.md

commit d57e9f42d974473789c569a10ea8ac10c3c92295
Author: Sunith <135949976+Sunith3210@users.noreply.github.com>
Date: Tue Aug 25 16:30:07 2026 +0530

    Create test1

commit cded1e18a1da8c7c766eef39e97ab63ac198bbdb
Author: sunith <sunithmalem@gmail.com>
Date: Tue Aug 25 16:24:23 2026 +0530

    this added in my local machine

HP@Ashish MINGW64 ~/Downloads/Git (master)
$ |
```

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22. Merge multiple commits into single commit.

git add.

git commit -m "messages"

git rebase -I HEAD~1

```
[detached HEAD 737a122] Create README.md
Author: Sunith <135949976+Sunith3210@users.noreply.github.com>
Date: Tue Aug 25 17:54:57 2026 +0530
3 files changed, 1 insertion(+)
create mode 100644 README.md
create mode 100644 a
create mode 100644 b
Successfully rebased and updated refs/heads/master.

HP@Ashish MINGW64 ~/Downloads/Git (master)
$ git log
commit 737a122449fdf1bd7991e30bbd1c3f91cbd73f8c (HEAD -> master)
Author: Sunith <135949976+Sunith3210@users.noreply.github.com>
Date: Tue Aug 25 17:54:57 2026 +0530

    Create README.md

    added in my local machine

    this line is added in local machine
```